

PhysicsDiff-PCB: Physics Guided Conditional Diffusion for Crosstalk Aware PCB Layout Generation

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Abstract—High-speed PCB design is increasingly challenged by electromagnetic interference effects, particularly near-end crosstalk between closely spaced transmission traces. This paper presents PhysicsDiff-PCB, a physics guided conditional diffusion framework for generating binary copper-mask PCB layouts directly from electrical and stack-up specifications. The proposed framework integrates a conditional diffusion architecture with physics-aware optimisation to produce layouts that are both geometrically coherent and electrically compliant. A frozen surrogate predictor is incorporated during training to provide differentiable guidance based on impedance and crosstalk behaviour, enabling the generator to align layout synthesis with underlying electromagnetic constraints. The training strategy combines conditional denoising diffusion, cross-attention conditioning, mask-aware supervision, and smoothness regularisation to improve structural consistency and manufacturability. Experimental results demonstrate that the proposed approach can generate physically meaningful PCB trace layouts while preserving signal integrity objectives, highlighting the potential of physics guided generative AI for automated high-speed PCB design and intelligent electronic layout synthesis.

Index Terms—Physics-guided diffusion models, inverse electromagnetic design, PCB layout synthesis, crosstalk mitigation, microstrip structures, differentiable physics constraints, DDPM, FDTD acceleration.

I. INTRODUCTION

At data rates exceeding 10 Gbps, characteristic impedance Z_0 and near end crosstalk (NEXT) become first order PCB design constraints whose satisfaction requires careful control of trace geometry [1]. The core challenge is *inverse design*: given target specifications Z_0^* and V_{NEXT}^* , synthesising a geometrically valid copper layout that meets both remains largely unautomated. Existing EDA tools enforce design rules reactively rather than generating geometry from performance targets, leaving engineers to close the loop iteratively through field solver feedback. ML surrogates accelerate electromagnetic forward prediction [2] and crosstalk characterisation [3], but are discriminative: surrogate inversion is ill posed and produces non-binary distributions unfit for fabrication [4]. Deterministic optimisation yields single point solutions and cannot represent the many geometrically distinct layouts that collectively satisfy a specification.

Conditional denoising diffusion probabilistic models [5] offer a principled alternative, learning $p(\mathbf{x} | \mathbf{c})$ over layouts given a specification vector and enabling diverse, structurally coherent generation via efficient DDIM sampling [6]. Prior work confirms diffusion for engineering inverse design [7],

[8], but naïve conditional diffusion provides no in loop electromagnetic feedback; integrating full-wave solvers at training scale is computationally intractable. We resolve this through *surrogate-guided training*: a frozen CNN PHYSICSPREDICTOR trained on Wadell impedance approximations [9] and coupled line crosstalk proxies [10] provides differentiable Z_0 and V_{NEXT} gradient signals without invoking a field solver.

We present **PhysicsDiff-PCB**, a surrogate-guided conditional diffusion framework that conditions a U-Net DDPM [11] on an 8-dimensional electromagnetic specification vector via cross-attention [12]. The composite training objective combines an SNR-weighted diffusion loss [13], Dice-Focal mask loss [14], physics-guided Huber losses propagated through the frozen PhysicsPredictor, and total variation regularisation [15], balanced using adaptive multi-task weighting [16]. To complement this surrogate-guided optimisation, representative generated layouts are converted into physical geometries and independently validated using the openEMS full-wave FDTD solver [17], assessing electromagnetic behaviour beyond the surrogate alone.

The principal contributions are: (i) specification-conditioned diffusion for inverse PCB layout synthesis; (ii) surrogate-guided physics feedback decoupled from full wave simulation; (iii) a fabrication aware composite loss with adaptive balancing; and (iv) ensemble-based pixel wise uncertainty quantification over generated layouts.

II. LITERATURE REVIEW

High speed PCB design under electromagnetic interference (EMI) constraints lies at the intersection of computational electromagnetics (CEM), physics informed machine learning (PIML), and deep generative modelling. Existing work has advanced forward electromagnetic prediction and physics constrained learning, but generating PCB trace layouts directly from electrical specifications remains underexplored.

A. Surrogate and Physics-Informed Modelling

ML surrogates have been widely used to approximate expensive electromagnetic simulations: Torun et al. [18] predicted PCB interconnect S-parameters with feedforward ANNs, and Kagawa et al. [19] extended this to CNN based near-field EMI prediction for microstrip structures over 1-3 GHz. Both remain discriminative and forward directed, returning an electrical response for a fixed geometry rather

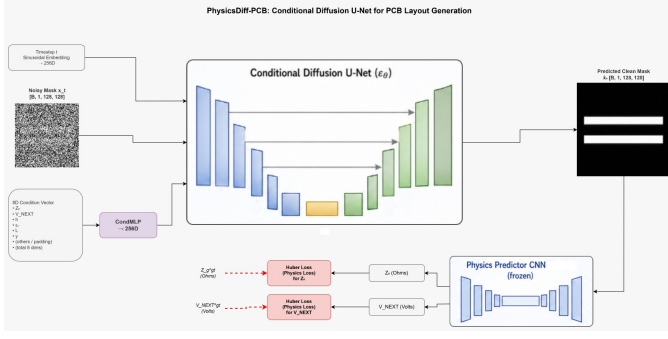


Fig. 1. Architecture of PhysicsDiff-PCB. A conditional DDPM U-Net predicts noise $\epsilon_\theta(x_t, t, c)$ from an 8-D EM specification vector and sinusoidal timestep embedding, with cross-attention confined to the 8×8 bottleneck. A frozen PhysicsSurrogateV2 provides impedance $\hat{Z}_0$ and crosstalk $\hat{V}_{\text{NEXT}}$ feedback via Huber losses. At inference, DDIM ($S=50, \eta=0$) generates compliant copper masks.

than generating geometry from a specification. PINNs [20] and hPINNs [21] instead embed governing-equation residuals into the training objective for hard constrained inverse design, but are deterministic and impractical to couple with large-batch diffusion training. PhysicsDiff-PCB addresses both gaps: it learns a conditional generative mapping from EM specifications to copper trace layouts, guided by a frozen CNN-based PhysicsPredictor trained on Wadell impedance [9] and coupled transmission line crosstalk [10] labels. Impedance and crosstalk errors back propagate through this surrogate during diffusion training without requiring solver calls, and the predictor’s fidelity can be upgraded independently by retraining on higher-quality simulation data [17].

B. Conditional Diffusion and Fabrication Aware Generation

DDPMs [5] offer stable training and strong sample diversity for stochastic inverse design, and DDIM [6] reduces inference cost by reinterpreting the reverse process as a deterministic ODE. Classifier free guidance [22] and ControlNet style conditioning [23] show that diffusion models can be steered by low dimensional external signals, an insight adopted in PhysicsDiff-PCB where an 8 dimensional EM specification vector is injected into the U-Net [24] via cross attention [12]. GANs have also shown strong performance in adjacent engineering applications, including electromagnetic signal denoising and enhancement for robust drone motor detection, underscoring the broader potential of adversarial learning for physics-related data generation [25]. Vlassis and Sun [7] confirmed generative inversion is viable for high dimensional engineering design, though without explicit physical constraints, precisely the gap PhysicsDiff-PCB closes. However, noise prediction loss alone is insufficient for PCB copper-mask generation, which requires sharp, binary, and continuous traces. PhysicsDiff-PCB therefore augments the base diffusion loss with a Dice and Focal loss [14] combination for foreground overlap and boundary accuracy, and a total variation regulariser [15] to suppress artefacts and promote trace continuity, with diffusion, mask, and physics terms adaptively

balanced following GradNorm [26] and uncertainty based multi task weighting [16]. Existing PCB EMI surrogates [18], [19] remain forward only and PINN methods [20], [21] cannot sample diverse designs, while prior diffusion-based inverse design [7] lacks embedded physical constraints; PhysicsDiff-PCB is the first framework to combine conditional diffusion-based layout generation, physics guided gradient feedback, and fabrication aware mask regularisation for high-speed PCB trace synthesis.

III. METHODOLOGY

A. Problem Formulation

PhysicsDiff-PCB is formulated as a surrogate guided conditional denoising diffusion framework for inverse PCB trace-layout generation. Rather than evaluating the electrical response of a fixed geometry, the model addresses the inverse problem: given electromagnetic and material specifications, it generates a copper mask layout expected to satisfy those specifications.

The generated layout is a single channel copper mask image, where copper pixels are close to one and substrate pixels are close to zero. The model learns a conditional distribution,

$$p_\theta(x|c),$$

where $x \in \mathbb{R}^{1 \times 128 \times 128}$ is the copper mask image and $c \in \mathbb{R}^8$ is the electromagnetic conditioning vector. The present implementation focuses on two-trace microstrip-like PCB structures.

B. Fast2D Surrogate Guided Dataset

A synthetic Fast2D dataset of 10,000 procedurally generated PCB layouts is used for training, sampled over physical parameters $[w, s, L, h, t, \epsilon_r, \tan \delta]$ (trace width, spacing, length, substrate height, copper thickness, permittivity, and loss tangent) using Latin Hypercube Sampling [27]. Invalid geometries ($w \geq s$ or $t \geq h$) are rejected. Each accepted sample is stored in HDF5 format [28] as a 256×256 three-channel layout tensor together with its physical parameters, impedance estimate, crosstalk proxy, and binary compliance label; full dataset generation details are given in Section IV-A. During training, only the copper channel is used, resized to 128×128 and normalised to $[-1, 1]$.

Analytical proxy labels are used in place of full-wave simulation for every sample: characteristic impedance Z_0 via a microstrip approximation [9] and near-end crosstalk V_{NEXT} via coupled-line transmission-line theory [10] (formulas given in Section IV-A). A binary compliance label is assigned using $|Z_0 - 85\Omega| \leq 10\Omega$ and $V_{\text{NEXT}} \leq 1.0$ V, the nominal impedance and crosstalk limits for this surrogate-guided stage.

C. Conditioning Vector

Each sample is represented by the 8-dimensional conditioning vector

$$c = [\hat{Z}_0, \hat{V}_{\text{NEXT}}, Z_0^*, V_{\text{NEXT}}^*, h, \epsilon_r, L, y],$$

where $\hat{Z}_0$ and $\hat{V}_{\text{NEXT}}$ are analytical proxy labels, $Z_0^* = 85\Omega$ and $V_{\text{NEXT}}^* = 1.0$ V are nominal design targets, and y is the

compliance label. The condition vector is standardised using training-set statistics, applied consistently across validation, testing, and inference.

D. Conditional Diffusion Model and U-Net Architecture

The generator follows the DDPM formulation [5]. During the forward process, Gaussian noise is added to a clean copper mask x_0 :

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

and the Conditional U-Net predicts the added noise,

$$L_{\text{diff}} = \|\epsilon - \epsilon_\theta(x_t, t, c)\|^2.$$

A cosine noise schedule is used, with DDIM sampling applied at inference to reduce reverse denoising steps [6].

The denoising backbone is a single-channel Conditional U-Net [24] operating on 128×128 masks, using residual blocks with Group Normalisation [29] and SiLU activations, with channel width increasing from 64 to 512 at the bottleneck. The diffusion timestep is encoded via sinusoidal embeddings, while the 8-dimensional condition vector is passed through a condition MLP to a 256-dimensional embedding and injected into deeper U-Net layers via cross-attention [12]; self-attention is also applied at the deepest resolution for global reasoning over the two-trace layout.

E. PhysicsPredictor Surrogate

A frozen CNN-based PhysicsPredictor (PhysicsSurrogateV2) provides differentiable electrical feedback during training, receiving the denoised copper-mask estimate and physical parameters and predicting impedance and near-end crosstalk:

$$(\hat{Z}_0, \hat{V}_{\text{NEXT}}) = f_{\text{phys}}(\hat{x}_0, a).$$

It is trained separately on the same Fast2D dataset using analytical proxy labels, with weights frozen during diffusion training, so physics-loss gradients update only the Conditional U-Net, enabling surrogate-guided feedback without a full-wave solver in the training loop.

F. Composite Training Objective

The total objective combines diffusion, mask, physics, and smoothness terms:

$$L_{\text{total}} = L_{\text{diff}} + w_{\text{mask}}L_{\text{mask}} + \lambda_{Z_0}L_{Z_0} + \lambda_{\text{XT}}L_{\text{XT}} + \lambda_{\text{sm}}L_{\text{sm}}.$$

The mask loss combines Dice and Focal loss for foreground overlap and boundary quality [14]; the physics terms compare PhysicsPredictor outputs against impedance and crosstalk targets via robust Huber losses; and a weak total variation term suppresses artefacts and promotes trace continuity [15]. Physics and smoothness losses are applied only at low-noise timesteps, where the denoised estimate is structurally meaningful.

G. Training and Inference

Training proceeds in two phases: the Conditional U-Net is first trained with the diffusion objective alone, then fine-tuned with the composite surrogate-guided loss once the frozen PhysicsPredictor is introduced. The dataset is split 80/10/10 using compliance-stratified sampling. AdamW [30] is used in PyTorch [31], with exponential moving average weights maintained for stable inference.

At inference, an 8-dimensional specification vector and Gaussian noise image are provided to the model; DDIM reverse sampling generates a soft copper mask, thresholded to obtain a binary layout. Multiple samples from the same condition (via different noise seeds) enable output diversity and uncertainty inspection.

H. Validation Scope

The framework is trained using the Fast2D surrogate-guided pipeline, where the frozen PhysicsPredictor provides differentiable feedback during optimisation. Reported surrogate-based metrics are complemented by representative independent full-wave validation using the openEMS FDTD solver [17]. While this confirms the physical feasibility of representative samples, large-scale FDTD benchmarking and experimental measurements remain future work.

IV. DATASET PLANING

A. Dataset Generation

A synthetic corpus of 10,000 two-trace microstrip geometries was generated as the Fast2D pilot dataset through a CPU-based multiprocessing pipeline.

Parameter Sampling: Seven continuous geometric and material parameters (Table I) were jointly sampled using Latin Hypercube Sampling (LHS, seed 42) for space-filling design-space coverage. A $3\times$ oversampling strategy (30,000 raw draws) was used to obtain exactly $N = 10,000$ valid configurations after rejecting geometrically invalid rows where $w \geq s$ or $t \geq h$ (trace width must be strictly less than spacing, and copper thickness strictly less than substrate height).

TABLE I
LHS PARAMETER SAMPLING RANGES

Symbol	Parameter	Min	Max
w	Trace width (mm)	0.075	0.500
s	Inter-trace spacing (mm)	0.100	0.800
L	Trace length (mm)	10.00	24.00
h	Substrate height (mm)	0.200	1.000
t	Copper thickness (mm)	0.018	0.070
ϵ_r	Relative permittivity	2.200	6.000
$\tan \delta$	Loss tangent	0.005	0.030

Layout Rasterisation: Each configuration was rasterised into a three-channel 256×256 image at $10 \mu\text{m}/\text{pixel}$ resolution with a 2 mm end margin: a binary copper occupancy mask (aggressor and victim trace rectangles, with aggressor centred at $y = w/2$ and victim at $y = w + s + w/2$, both spanning length L), a uniform material channel (reserved for future dielectric contrast), and a normalised substrate height channel $\min(1, h/1.0)$. Only the copper channel is used during training

(Section III). Layout tensors were saved as compressed .npz arrays with .png previews.

Analytical EM Label Computation: Characteristic impedance was estimated using the Wadell microstrip formula:

$$\hat{Z}_0 = \frac{87}{\sqrt{\epsilon_r + 1.41}} \ln \left(\frac{5.98 h}{0.8 w + t} \right) [\Omega], \quad (1)$$

and near-end crosstalk via a lumped coupled-transmission-line (CTL) proxy capturing exponential decay with normalised spacing and dependence on length, permittivity, and width:

$$\hat{V}_{\text{NEXT}} = e^{-1.6 s/h} \cdot \frac{L}{20} \cdot \frac{\epsilon_r}{4.5} \cdot \frac{0.35}{w}. \quad (2)$$

A binary compliance label was assigned per sample:

$$\ell = \begin{cases} 1 & \text{if } |\hat{Z}_0 - 85| \leq 10 \Omega \text{ and } \hat{V}_{\text{NEXT}} \leq 1.0 \text{ V,} \\ 0 & \text{otherwise.} \end{cases} \quad (3)$$

A synthetic Gaussian proxy field map, modelling the near-field envelope as a spacing-dependent superposition of aggressor and victim Gaussian cross-sections, was also generated as a spatially resolved supervision signal complementary to the scalar labels.

Storage and Splits: All artefacts were packed into a single gzip-compressed HDF5 file (layout_3ch, field_topview_fast, params, z0_fast, next_fast, label_compliant) and split 80/10/10 into training, validation, and test subsets using compliance-stratified sampling.

V. EXPERIMENT AND RESULTS

A. Experimental Environment

Dataset generation ran on CPU with Python multiprocessing; model training used GPU acceleration with PyTorch DataParallel where multiple devices were available. The full pipeline – DDPM schedule, DDIM sampler, U-Net backbone, PhysicsPredictor, and composite loss functions – was implemented from scratch in PyTorch, enabling full control over physics-loss integration and adaptive scheduling.

B. Training Setup

The complete training configuration is summarised in Table II. The Conditional U-Net was trained for 200 epochs with mixed-precision (`torch.amp`) and gradient clipping at 1.0 to maintain stability during the physics loss ramp. An EMA of all parameters (decay 0.9999) was maintained and used exclusively for inference and validation, as EMA smoothing consistently outperforms raw checkpoint weights; the best validation-loss checkpoint was retained separately to support ablation studies. The PhysicsPredictor was trained separately on the same Fast2D corpus, mapping copper masks and material vectors to normalised ($\hat{Z}_0$, $\hat{V}_{\text{NEXT}}$) labels, and its weights were frozen prior to and throughout diffusion training.

TABLE II
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Optimiser	AdamW ($\text{lr}=2 \times 10^{-4}$, $\text{wd} = 10^{-4}$)
LR Schedule	CosineAnnealingLR ($T_{\text{max}}=200$, $\eta_{\text{min}}=10^{-6}$)
Epochs	200
Batch Size	32 (single-GPU); scaled w/ DataParallel
Diffusion Steps T	1,000 (cosine β -schedule)
DDIM Steps	50
Train Res.	128×128 (prod. 256×256)
EMA Decay	0.9999
Mixed Precision	<code>torch.amp</code> , <code>grad clip</code> = 1.0
Physics Warmup	Ep. 0–9: $\lambda_{Z_0} = \lambda_{XT} = \lambda_{\text{smooth}} = 0$
Physics Ramp	Ep. 10–24: linear anneal to adaptive
EMA Adapt. Decay	0.98; $\min \lambda_{Z_0} = 0.01$
Weights	$\min \lambda_{XT} = \min \lambda_{\text{smooth}} = 0.005$
w_{mask}	0.15 (fixed, Dice+Focal)
Ensemble (UQ)	$N=50$ independent DDIM trajectories

C. Comparison of Models

TABLE III
COMPARISON OF PCB LAYOUT GENERATION MODELS. BEST VALUES ARE SHOWN IN BOLD.

Metric	PhysicsDiff	CGAN	VAE	Uncond.	Cond.
IoU $\uparrow$	0.612	0.737	0.815	0.347	0.087
Dice $\uparrow$	0.754	0.833	0.905	0.486	0.159
SSIM $\uparrow$	0.718	–	–	0.458	0.456
gray!10 Z_0 Err. (Ω) $\downarrow$	5.21	23.44	17.83	5.01	22.76
gray!10 NEXT Err. (V) $\downarrow$	0.183	0.776	0.744	0.804	0.725
gray!10 Compliance $\uparrow$	0.578	0.270	0.344	1.000	0.259
Cond. Resp. $\uparrow$	0.312	0.001	0.112	–	0.074
Struct. Gap $\downarrow$	0.083	0.007	0.008	1.214	0.587
Diversity $\uparrow$	0.041	0.0003	0.175	0.057	0.055

Four baselines were evaluated alongside PhysicsDiff-PCB: PhysicsCGAN, PhysicsVAE, an unconditional DDPM, and a conditional DDPM. PhysicsDiff-PCB provides the best overall trade-off between geometric fidelity, electrical accuracy, and conditioning capability, achieving the lowest NEXT error (0.183 V) and highest conditioning responsiveness (0.312), confirming that surrogate-guided physics optimisation aligns generated layouts with electromagnetic specifications while preserving structural consistency.

PhysicsCGAN reconstructs competitive trace patterns via adversarial learning but shows higher impedance/crosstalk errors and lower compliance, indicating adversarial optimisation alone is less effective at enforcing electrical constraints. PhysicsVAE achieves the highest IoU/Dice from its latent-space reconstruction objective, yet its electrical performance and conditioning remain inferior – high geometric similarity does not guarantee electrical compliance. Among the diffusion-only baselines, the unconditional DDPM’s apparently strong impedance and compliance reflect a degenerate low-fill solution rather than meaningful trace generation, while the conditional DDPM collapses under cross-attention alone, with poor conditioning responsiveness and weak structural quality. These results confirm that the surrogate-guided physics losses $\mathcal{L}_{Z_0}$ and $\mathcal{L}_{XT}$ are essential for stable, controllable, electrically consistent generation.

D. Qualitative Generation Results

Figs. 2 and 3 compare compliant and non-compliant generated layouts from the held-out test set. Both recover continuous

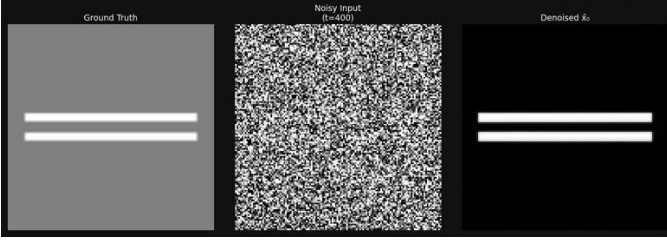


Fig. 2. Representative surrogate-compliant generated layout. The generated mask is visually trace-like and satisfies the surrogate-predicted impedance and crosstalk limits.

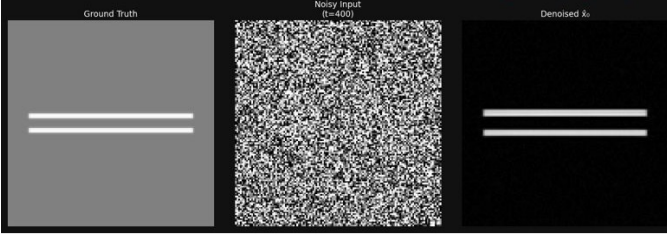


Fig. 3. Representative surrogate non compliant generated layout. Although the output remains visually trace like, it fails the surrogate-based electrical compliance check.

two-trace structures after DDIM denoising and binary thresholding, but only Fig. 2 satisfies the surrogate-based electrical criteria – confirming that visual quality alone is insufficient for PCB validation.

E. Evaluation Methodology

Model quality was assessed along four complementary axes: **training convergence**, tracking composite and component losses and adaptive weights λ_{Z_0} , λ_{XT} , λ_{smooth} across all 200 epochs to confirm warmup/ramp schedules and EMA-based normalisation; **qualitative layout fidelity**, comparing binarised (0.5 threshold) generated masks against held-out ground truth for trace continuity, boundary sharpness, and conditioning consistency; **physics validation**, passing binarised masks through the frozen PhysicsPredictor to extract $\hat{Z}_0$ and $\hat{V}_{NEXT}$ and comparing against targets via MAPE, with a target of crosstalk prediction within 15% of FDTD ground truth and compliance rate reported as a proxy-model baseline; and **uncertainty quantification**, sampling $N = 50$ independent DDIM trajectories per test condition and computing pixel-wise ensemble variance, flagging compliant layouts with high variance as uncertain.

F. Ablation: Label Leakage in the Conditioning Vector

To verify that dimensions 0 ($\hat{Z}_0$), 1 ($\hat{V}_{NEXT}$), and 7 (y) do not leak answer labels into generation, we compared a **5D Ablated** variant ($c = [Z_0^*, V_{NEXT}^*, h, \epsilon_r, L]$, leaky dims removed) against an **8D** variant (full vector, dims 0,1 set to target at inference) using a reduced proxy model (16-ch U-Net, $T=150$, 500 samples, 50 epochs). Results are shown in

Table IV. The 5D model achieves lower Z_0 error (34.24% vs. 41.79%), confirming that removing leaky dims does not degrade primary impedance quality. Zero compliance in both variants reflects the reduced training budget, not a conditioning failure.

TABLE IV
LABEL LEAKAGE ABLATION (PROXY MODEL)

Variant	Z_0 MAPE	Z_0 MAE	VN MAPE	Comply
5D Ablated	34.24%	29.1 Ω	107.45%	0%
8D	41.79%	35.5 Ω	80.05%	0%

G. Independent openEMS Validation

To independently assess the electromagnetic behaviour of the generated layouts, representative masks produced by PhysicsDiff-PCB were exported to the openEMS full-wave FDTD simulator [17]. Each binary copper mask was converted into its physical geometry by extracting trace width (w), spacing (s), and length (L) directly from the layout, then used to automatically construct an equivalent microstrip transmission-line model in openEMS with the same substrate height and dielectric properties used during dataset generation. Unlike the surrogate-guided framework, openEMS performs a full-wave FDTD simulation without the frozen PhysicsPredictor; characteristic impedance (Z_0), near-end crosstalk (V_{NEXT}), return loss (S_{11}), and insertion loss (S_{21}) are extracted from the simulated S-parameters.

TABLE V
REPRESENTATIVE COMPARISON BETWEEN TARGET SPECIFICATIONS, PHYSICSPREDICTOR ESTIMATES, AND INDEPENDENT OPENEMS FDTD RESULTS.

Metric	Target	PhysicsPredictor	openEMS FDTD
Z_0 (Ω)	112.38	116.52	112.36
V_{NEXT}	0.1596	0.2006	0.2663

As shown in Table V, the PhysicsPredictor’s impedance estimate differs only slightly from the independent openEMS result, indicating the surrogate preserves the layout’s overall electromagnetic behaviour. A larger discrepancy is observed for near-end crosstalk, reflecting the greater complexity of modelling coupled electromagnetic interactions with analytical proxy labels. Nevertheless, the openEMS results confirm that PhysicsDiff-PCB generates physically meaningful layouts independently verifiable by a full-wave solver, complementing the surrogate-based validation above.

VI. DISCUSSION

PhysicsDiff-PCB demonstrates that integrating physics-aware constraints into a conditional diffusion model enables rapid, layout-consistent PCB trace synthesis without the computational overhead of full-wave electromagnetic solvers. By conditioning on an eight-dimensional parameter vector, the model learns a physically anchored latent representation of PCB geometry that purely data-driven methods cannot reliably provide. The frozen PhysicsPredictor injects electromagnetic

consistency into the training objective without backpropagation through expensive numerical solvers, decoupling geometric learning from physical plausibility enforcement.

The Latin Hypercube sampled synthetic dataset substantially reduces preparation time and enables rapid iteration; however, the analytical approximations are valid primarily under quasi-static assumptions and may underestimate higher-frequency effects, while structural field maps capture geometric topology rather than physically solved near-field distributions, defining the current accuracy ceiling. Independent validation of representative layouts using the OpenEMS full-wave FDTD solver [17] confirms physically meaningful electromagnetic behaviour while also exposing the surrogate’s remaining limitations, particularly for near-end crosstalk prediction. A label leakage ablation (Table IV) further confirmed that the 5D conditioning formulation matches or outperforms the full 8D variant on the primary impedance metric (Z_0 MAE: 29.1 vs. 35.5 Ω), validating the design against leakage concerns.

The modular conditioning and loss formulation allow dataset substitution without architectural redesign, preserving scalability toward solver-grounded data and supporting broader goals of accelerating inverse PCB design workflows and enabling EMI-aware layout automation at the pre-route stage.

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